Proof. Let $\alpha = \mathrm{rad}(\mathfrak{a})$. Then, for any $a \in \mathfrak{a}$, b is algebraic over (a, α) and hence, by Lemma 2.13, is algebraic over a. Since there is an $a \in \mathfrak{a}$ such that $a_i \notin \mathrm{acl}(a_{\tilde{i}}, b)$ for all $i \in [n]$, by Lemma 2.14, b must be algebraic. □

Lemma 2.19 ([16, Lemma 4.15])**.** *If $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated then every algebraic (respectively, definable) closed disc contains an algebraic (respectively, definable) point.*

Lemma 2.20 ([16, Lemma 4.17])**.** *If $\mathbb{S}$ is (VF, Γ)-generated then every algebraic (respectively, definable) disc contains an algebraic (respectively, definable) point.*

A pillar of the structure of definable sets in $\mathbb{U}$ is *C-minimality*, meaning that every definable subset of VF is a finite boolean combination of valuative discs, allowing additional parameters to name the individual discs.

Resolved referee R027 [Clarification] – Parameters of the balls in C-minimality.
Original source line 859. The balls in a finite Boolean decomposition need not be individually $\mathbb{S}$-definable. For example, a definable pair of conjugate algebraic points need not split into two definable singleton balls. Say 'balls with additional parameters (whose finite canonical family is algebraic over the defining parameters)' and retain the Galois-invariant family where individual components cannot be chosen.

Author resolution. Unqualified definability remains relative to the fixed ground parameter set. The individual discs in a finite Boolean decomposition may require additional parameters to be named; the sentence now makes that distinction explicit.

Corollary 2.21. *Suppose that $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated. Let $\mathfrak{a} \subseteq \mathrm{VF}$ be a definable disc and A a definable subset of* VF*. If $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable point.*

Proof. It is not hard to see that, by C-minimality, if $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable closed disc and hence the claim is immediate by Lemma 2.19. □

Definition 2.22. Let D be a set of parameters. We say that a (not necessarily definable) nonempty set A *generates a (complete) D-type* if, for every D-definable set B, either $A \subseteq B$ or $A \cap B = \emptyset$. In that case, A is *D-type-definable* if no set properly contains A and also generates a D-type. If A is D-definable and generates a D-type, or equivalently, if A is both D-definable and D-type-definable then we say that A is *D-atomic* or *atomic over D*.

We simply say "atomic" when $D = \emptyset$.

In the literature, a type could be a partial type and hence a type-definable set may have nontrivial intersection with a definable set. In this paper, since partial types do not play a role, we shall not carry the superfluous qualifier "complete" in our terminology.

Lemma 2.23 ([18, Lemma 3.3])**.** *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let A be an atomic set. Then A is γ-atomic for all $\gamma \in \Gamma$.*

Resolved referee R028 [Error] – Atomic base change needs a nontrivial base value group.
Original source line 879. The unrestricted statement remains false over a trivially valued base, as the human referee observed. Over a trivially valued algebraically closed field, $A = \mathcal{M} \setminus \{0\}$ is atomic; a parameter $\gamma > 0$ cuts it into the nonempty proper subset $\{x : 0 < \mathrm{val}(x) < \gamma\}$. Impose a nontrivial base value group here and propagate that assumption through uses of this lemma. In the cited proof, the essential step is that a nonempty definable subset of Γ contains a definable point; this fails for $(0, \infty)$ over the trivial base.

Author resolution. The author adopts nontrivial $\Gamma(\mathbb{S})$ as a standing assumption throughout the paper, matching the blanket hypothesis in [18]. It persists under parameter extension and supplies the

required hypothesis here and in subsequent uses, including Lemma 2.25 and Corollary 2.30. The contrary permission at the opening of Section 6 is removed.

Lemma 2.24. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc. Then $\mathfrak{a}$ is t-atomic for all $t \in \mathrm{RV}$.*

Proof. For the case $n = 1$, if $\mathfrak{a}$ were not t-atomic then, by Corollary 2.21 and Lemma 2.13, it would contain a point algebraic over the fixed ground base. This point belongs to a finite ground-definable set. Atomicity would then force the whole infinite open disc $\mathfrak{a}$ to lie in that finite set, a contradiction. For the case $n > 1$, by induction on n, $\mathfrak{a}_1$ is t-atomic and for every $a \in \mathfrak{a}_1$, $\mathfrak{a}_{>1}$ is (a, t)-atomic. So $\mathfrak{a}$ cannot have a nonempty proper t-definable subset. □

Partly resolved referee R029 [Gap] – Atomicity proof: algebraic versus definable.
Original source line 886. Lemma 2.13 only gives an algebraic point over the original base, not necessarily a definable point. This suffices: an infinite atomic disc cannot meet a finite base-definable set. Replace 'definable point' at this step by that argument. The final sentence also needs 'nonempty proper t-definable subset'. Finally, indicate why Corollary 2.21 remains applicable whenever previously named disc parameters occur in the base; its stated hypothesis excludes such extra parameters.

Author resolution. The author approves both proof corrections: the point is algebraic over the fixed ground base, so atomicity would place the infinite disc inside a finite ground-definable set; and the final sentence now says "nonempty proper." The statement is unchanged. The separate question about applicability over additional disc-imaginary ground parameters remains open.

Lemma 2.25. *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let $A \subseteq \mathrm{VF}^n$ be a set that generates a type and $\mathfrak{b} \subseteq A$ an open (or closed) polydisc. Then $\mathfrak{b}$ is $\ulcorner \mathfrak{b} \urcorner$-atomic.*

Proof. We do induction on n. The base case $n = 1$ is just [18, Lemma 3.4]. For the case $n > 1$, write $\mathfrak{b} = \prod_i \mathfrak{b}_i$, $\mathfrak{b}' = \prod_{i>1} \mathfrak{b}_i$, and $\gamma = \mathrm{rad}(\mathfrak{b}_1)$. For any $a \in \mathfrak{b}_1$, A_a generates a type. So, by Lemma 2.23 and the inductive hypothesis, $\mathfrak{b}'$ is $(a, \gamma, \ulcorner \mathfrak{b}' \urcorner)$-atomic and hence $\ulcorner \mathfrak{b} \urcorner$-atomic. Similarly $\mathfrak{b}_1$ is $\ulcorner \mathfrak{b} \urcorner$-atomic. It follows that $\mathfrak{b}$ must be $\ulcorner \mathfrak{b} \urcorner$-atomic. □

Lemma 2.26. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc and $f : \mathfrak{a} \longrightarrow \mathcal{P}(\mathrm{VF})$ a definable finitary function such that $\mathrm{rv} \restriction f(a)$ is injective for every $a \in \mathfrak{a}$. Then $\mathrm{rv} \circ f$ is constant.*

Proof. This is clear since, by Lemma 2.24, for every $t \in \mathrm{rv}(\bigcup f(\mathfrak{a}))$, $\mathfrak{a}$ is t-atomic and hence f induces a t-definable function $\mathfrak{a} \longrightarrow t^\sharp$. The lemma follows. □

A *thin annulus* is a set of the form $\mathfrak{c}(a, \gamma) \setminus \mathfrak{o}(a, \gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x - a) = \gamma\}$, where $a \in \mathrm{VF}$ and $\gamma \in \Gamma$. Equivalently, it is a closed disc with one maximal open subdisc removed.

Lemma 2.27 ([18, Lemma 3.8])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc or an atomic closed disc or an atomic thin annulus. Let $f : A \longrightarrow \mathrm{VF}$ be a definable function. Then $f(A)$ is also atomic of one of these three forms.*

Resolved referee R163 [Clarification] – Define thin annuli before using them.
Original source line 907. The term "thin annulus" has not been defined in this manuscript. Specify a closed disc with one maximal open subdisc removed; after choosing the center of the removed subdisc, it is a shell $\{x : \mathrm{val}(x - a) = \gamma\}$. This makes the alternatives here and the later component argument unambiguous.

Author resolution. The author approves the finite-radius shell definition $\mathfrak{c}(a, \gamma) \setminus \mathfrak{o}(a, \gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x - a) = \gamma\}$, with $\gamma \in \Gamma$. It is now stated before the first use in Lemma 2.27.